

# CASM: A CONTEXT-AWARE SEMI-MARKOV ALTERNATIVE TO DBN POST-PROCESSOR FOR BEAT TRACKING

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## ABSTRACT

The last stage of beat tracking must turn dense neural activations into a discrete, musically coherent event sequence. Direct peak picking preserves local evidence but can fragment the pulse, whereas dynamic Bayesian networks (DBNs) restore continuity through globally configured tempo and meter assumptions. PLPDP showed that these assumptions can instead be informed by local periodicity, but a locally preferred pulse may itself be ambiguous between competing metrical levels. Building on that insight, we introduce CASM, a plug-in, context-aware semi-Markov decoder over a sparse graph of activation-supported beat candidates. CASM measures the margin between the best and competing local periods and uses it to control both the strength and tolerance of each segment-duration potential. It commits when one pulse is well supported and defers to the neural evidence when several interpretations remain plausible. Beat-count and downbeat-agreement safeguards make this behaviour deterministic and auditable. One globally calibrated configuration is applied without backbone retraining or track-specific BPM and meter settings to Beat This, MSCNN-lite, and TCN activations on GTZAN and SMC. The results position CASM as a conservative F1-continuity operating point rather than a universal replacement for DBN.

Index Terms— beat tracking, downbeat, content-aware adaptation, semi-Markov decoding, post-processing, local periodicity

## 1. INTRODUCTION

The final decoder in a beat tracker is deceptively consequential. A neural network can place high framewise probability on genuine beats and still leave duplicates, gaps, metrical-level switches, or downbeats that disagree with its beat grid. Event F-measure rewards local placement, whereas continuity metrics also require one metrical interpretation to persist. Weak decoding therefore respects the network but may yield an unstable clock; strong decoding restores a clock but may overwrite correct, expressive, or unusual evidence.

A dynamic Bayesian network (DBN) addresses this problem by asking which path through tempo, phase, and meter states best explains all frames [1]. In plain terms, the input chooses a route through a globally specified timetable: the BPM range, allowed meters, and cost of tempo changes are fixed before the track is seen. The inferred state path is input-dependent, but the transition law does not become

softer merely because the activation reveals a credible half- or double-tempo alternative. CASM takes the complementary view. It considers only activation-supported peaks, scores the complete elapsed interval between two peaks, and lets the local activation decide how strongly regularity should count. Thus the practical difference is not “adaptive versus non-adaptive inference,” but a fixed DBN transition law versus an input-conditioned segment penalty. This distinction matters on difficult music, where retuning a DBN can help one corpus but weakens the claim of out-of-the-box transfer [2]. Conversely, post-processing-free trackers avoid some DBN errors but minimal peak picking can sacrifice sequence continuity [3, 4].

PLPDP provides the immediate starting point for our approach. It showed that a post-processor can replace a single global tempo preference with a local inter-beat interval and confidence derived from predominant local pulse (PLP) [5]. We retain its central idea—the temporal prior should listen to the current activation—and address a remaining risk: the strongest local period may only narrowly beat a competing metrical level. CASM therefore measures a best-versus-runner-up period margin. A clear winner creates a narrow, strong duration preference; a small margin weakens and broadens it, allowing the neural peaks to decide.

Fig. 1 previews these two behaviours on real activations. On the left, Direct retains mainly every second strong peak; CASM uses supported local periodicity to select intervening weak maxima and repairs continuity without moving events away from the activation. On the right, octave-related period hypotheses compete, so CASM suppresses its structural preference and leaves the Direct path unchanged. The latter occurs without invoking the count-ratio safeguard: uncertainty changes the segment score itself.

Our contributions are threefold:

- Building on PLPDP, we formulate beat decoding as a sparse semi-Markov path whose segment target, strength, and tolerance are determined by local period evidence and its ambiguity at both endpoints.
- We add beat-count and downbeat-agreement safeguards to obtain a deterministic plug-in that requires neither upstream retraining nor per-track BPM and meter settings.
- We study one frozen operating point across three activation backbones and analyse how decoder-calibration scale and composition affect transfer.

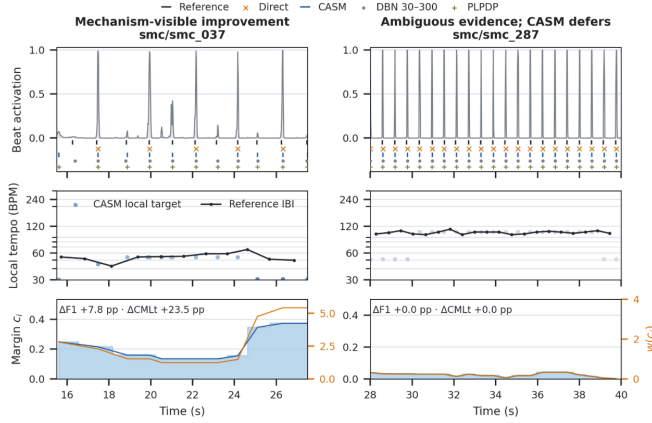


Fig. 1. Illustrative CASM behaviour on real Beat This OOF activations from SMC. Left: a supported local period helps recover weak, activation-supported beats. Right: a small period margin weakens the duration term, so CASM defers to Direct. Reference IBI is shown only for interpretation and is never observed by CASM. Tracks and windows were selected post hoc to expose the mechanism, not to estimate average performance.

## 2. RELATED WORK

### 2.1. Structured post-processing

Dynamic programming has long balanced onset evidence against inter-beat regularity [6]. Bayesian beat-position models used a dominant beat interval [7], while learned CRFs were later extended from beats to downbeats, long-range section consistency, and genre-specific microtiming [8–11]. Resonating comb filters estimate a dominant periodicity before event alignment [12], and DBNs enumerate tempo–phase–meter states and decode their most likely trajectory [1, 13]. These methods differ in state representation and optimization, but they share a useful principle: noisy framewise evidence should be interpreted as a sequence rather than thresholded independently.

### 2.2. From PLPDP to ambiguity-aware local constraints

PLPDP is CASM’s direct methodological predecessor [5]. Its central diagnosis was that a fixed global tempo penalty, including the empirically chosen transition law of common HMM trackers, can be too rigid for expressive timing. PLPDP extracts a time-varying inter-beat interval and a confidence curve from PLP, then replaces the global target and weight of dense-frame dynamic programming with those local quantities. Its real-time continuation improves controllability [14], and dPLP makes local-pulse estimation differentiable [15].

CASM advances this line of work rather than treating PLPDP as an unrelated baseline. PLPDP asks, “What is the current local pulse, and how strong is its PLP peak?” CASM additionally asks, “How decisively does that pulse beat the next plausible explanation?” It measures a top-versus-runner-up period margin around each retained activation maximum, combines the two endpoint contexts, and uses the result to change both the strength and the tolerance of

a complete event-to-event duration cost. The path is sparse and remains anchored to observed maxima, whereas PLPDP can evaluate every frame. This extension is designed for half/double-tempo competition: when no period wins clearly, the decoder should not enforce any of them confidently.

Reliability-informed tracking previously established the broader principle that temporal commitment should depend on evidence quality [16]. CASM’s margin is a local, deterministic separation score rather than predicted track-level accuracy. It also differs from the causal one-dimensional semi-Markov state space of [17], which maintains a frame-wise posterior and adapts a jump-back distribution online. CASM is an offline maximum-score path over candidate-to-candidate segments. Nor is it a semi-CRF: its potentials are hand-specified, unnormalized, and not learned [18].

### 2.3. Alternatives beyond plug-in decoding

Particle filters maintain online tempo hypotheses [19], and BeatNet/BeatNet+ extend this branch to joint real-time rhythm analysis [20, 21]. Another line adapts a tracker with user annotations [22–26]. More recent systems remove or redesign post-processing: masked diffusion iteratively refines coherent sequences [27], BeatFCOS predicts intervals as objects with Soft-NMS [28], and other work reformulates the prediction target [29]. These approaches change the predictor or import additional supervision. CASM instead addresses the deployment setting in which only the activation output of a frozen tracker is available.

## 3. METHODOLOGY

### 3.1. Overview and PLPDP lineage

Let a frozen tracker produce beat and downbeat logits  $z_t^b$  and  $z_t^d$  at frame rate  $F$ . Like PLPDP, CASM derives a local period and a measure of its support from the current beat activation. CASM then makes two changes. First, it decodes only a sparse set of activation maxima rather than every frame. Second, its support measure is the separation between the best and competing periods, and it controls both how strongly and how narrowly the chosen period is enforced. The global coefficients define one frozen evidence-to-constraint rule; the input determines the operating point of that rule at every candidate segment.

### 3.2. Candidate-event segments

Set  $a_t = \text{sigmoid}(z_t^b)$  and retain its local maxima above threshold  $\theta_c$  as  $\mathcal{C} = \{t_1, \dots, t_N\}$ . An edge  $(i, j)$  spans the complete elapsed time  $\Delta_{ij} = (t_j - t_i)/F$  and is allowed only within the 30–300 BPM support. With  $e_i$  denoting clipped activation evidence, CASM selects a path  $\pi = (i_1, \dots, i_K)$  that maximizes

$$\mathcal{S}(\pi) = \sum_{k=1}^K e_{i_k} - \sum_{k=2}^K D_{i_{k-1}, i_k}, \quad (1)$$

where  $D_{ij}$  is the segment-duration cost below. This is the precise sense in which CASM is semi-Markov: one transition consumes a variable-length candidate-to-candidate segment and scores its duration as a whole. It is not a generative hidden semi-Markov model, and the segment potentials are not learned.

### 3.3. Local period and ambiguity

At each candidate  $i$ , normalized autocorrelation scores the admissible integer lags  $\ell$  in a centred local window  $\mathcal{W}_i$ :

$$q_i(\ell) = \frac{\langle a_u a_{u-\ell} \rangle_{u \in \mathcal{W}_i}}{\sqrt{\langle a_u^2 \rangle_{u \in \mathcal{W}_i} \langle a_{u-\ell}^2 \rangle_{u \in \mathcal{W}_i}} + \epsilon_q} \left( \frac{\ell_{\min}}{\ell} \right)^\beta, \quad (2)$$

with zero padding outside the signal and a mild fast-tempo prior in the final factor. The winning lag gives  $\ell_i^* = \arg \max_{\ell} q_i(\ell)$  and local period  $\tau_i = \ell_i^* / F$ . We then compare it with the best lag outside a  $\pm 2$ -frame neighbourhood:

$$c_i = \text{clip}_{[0,1]} \left( \frac{q_i(\ell_i^*) - \max_{\ell: |\ell - \ell_i^*| > 2} q_i(\ell)}{|q_i(\ell_i^*)| + \epsilon_c} \right). \quad (3)$$

Here  $c_i$  is a period-separation margin, not a calibrated probability of being correct. A large value means that one local period clearly wins; a small value means that nearby or half/double-time explanations remain competitive. Both  $\tau_i$  and  $c_i$  are median-filtered over five consecutive candidates.

### 3.4. Ambiguity-conditioned segment cost

For edge  $(i, j)$ , CASM symmetrizes the endpoint contexts as  $\bar{\tau}_{ij} = \sqrt{\tau_i \tau_j}$  and  $c_{ij} = \sqrt{c_i c_j}$ . Define

$$\begin{aligned} \sigma(c) &= \sigma_0 + (1 - c)\sigma_u, \\ w(c) &= \frac{\lambda c}{2\sigma(c)^2}, \\ D_{ij} &= w(c_{ij}) \left[ \log \left( \frac{\Delta_{ij}}{\bar{\tau}_{ij}} \right) \right]^2. \end{aligned} \quad (4)$$

The log ratio treats proportional timing errors symmetrically. More importantly, the margin acts twice: high  $c$  narrows  $\sigma(c)$  and increases  $w(c)$ , while low  $c$  broadens the tolerated deviation and reduces its weight. CASM can therefore select weak but rhythmically supported maxima when the pulse is clear, yet avoid imposing a dubious grid when periodic evidence is ambiguous.

Fig. 2 connects Eq. (4) to the observed decoder behaviour. Panel (a) is the single response law fixed by the global coefficients. Panel (b) shows that real edges from different backbones and corpora occupy different parts of that law. Panel (c) aggregates the same quantity per piece: one frozen configuration produces distinct effective stiffness distributions, while the separately annotated fallback remains a second safety mechanism. These panels audit input conditioning; they are not an accuracy comparison.

### 3.5. Decoding and safeguards

A Viterbi-style dynamic program maximizes Eq. (1), with a restart option for an unfavourable prefix, and backtracks the highest-scoring path. In parallel, we form the Direct path of [4] by seven-frame max pooling, a zero-logit threshold, and one-frame deduplication. If the structured-to-Direct event-count ratio falls outside  $[r_{\min}, r_{\max}]$ , CASM returns Direct. This is a deterministic sanity check, not a learned router.

Downbeats are decoded only on the selected beat grid. A second dynamic program links candidate downbeats separated by an allowed bar length and penalizes unnecessary

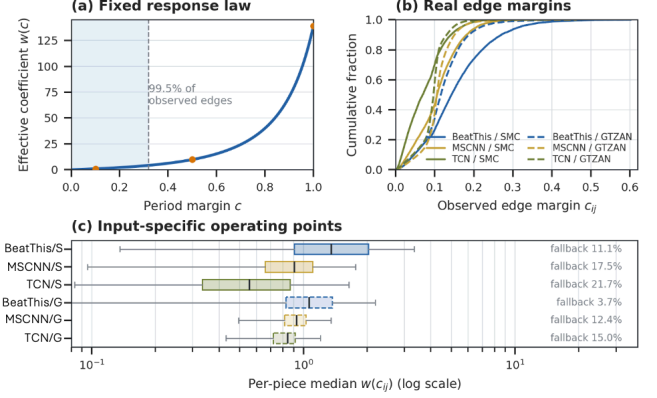


Fig. 2. Input-conditioned duration stiffness under one frozen CASM configuration. (a) The fixed constants map period margin  $c$  to coefficient  $w(c)$ ; shading contains 99.5% of observed structured-path edges. (b) Edge margin distributions differ across backbones and corpora. (c) Per-piece median coefficients therefore occupy different operating ranges; annotations give the separate count-ratio fallback rate. Fixed parameters define one response law, not one effective rigidity for every input.

meter changes. Direct downbeat maxima are independently snapped to the same beat grid; the meter path is retained only when it agrees sufficiently with that snapped path, otherwise the snapped path is returned. Thus every output downbeat remains synchronized with an output beat.

All scalar settings are calibrated once on development folds and then frozen for every track and backbone, as detailed in Sec. 4. CASM is therefore input-conditioned but not parameter-free: it has no learned weights, per-track tuning, annotated beats, BPM, or meter input at inference.

## 4. EXPERIMENTS

### 4.1. Data, backbones, and metrics

We evaluate on GTZAN ( $N = 993$ ) using the annotations and exclusions of Beat This [4], and on the deliberately difficult SMC corpus ( $N = 217$ ) [30]; SMC provides beat but not downbeat annotations. We use three 50-Hz activation backbones: the Beat This spectrogram Transformer [4], the lightweight multi-scale convolutional backbone MSCNN-lite [31], and the TCN of [32]. The decoder receives only their framewise beat and downbeat logits.

For GTZAN, final0, final1, and final2 are three independently trained full-data backbone seeds. Their training excludes GTZAN but includes SMC. We first macro-average over the same 993 tracks and then average the three seed-level scores. For SMC, using those full-data checkpoints would leak training examples; we instead use the corresponding held-out checkpoint for each of eight folds. Seven folds contain 27 tracks and the eighth contains 28; we average their fold-level macro scores. CASM itself is deterministic and has neither a neural checkpoint nor a random seed.

Following the established protocol, the first 5s are trimmed and mir\_eval computes F1 with  $\pm 70$  ms tolerance, CMLt, and AMLt [33]. CMLt requires a continuously correct

metrical level; AMLt additionally accepts the standard off-beat and half/double-level interpretations. Seed variation on GTZAN and fold variation on SMC have different replication units and are descriptive, not confidence intervals.

#### 4.2. Decoder baselines

Direct is the minimal peak picker defined in Sec. 3. DBN is the conventional joint madmom configuration: meters  $\{3, 4\}$ , 55–215 BPM, transition weight 100, observation weight 16, and threshold 0.05 [1]. DBN-SMC-opt is deliberately diagnostic. Its sequential beat and bar stages were selected from 753 configurations on the TCN SMC 8-fold OOF activations, maximizing beat F1 with CMLt/AMLt tie-breaks, and only then frozen for the three backbones and GTZAN. It uses 50–215 BPM, 30 tempo states, transition/observation weights 7/20, threshold 0.3, meters  $\{3, 4\}$ , bar-observation weight 100, and meter-switch probability  $10^{-7}$ . It is neither a clean SMC test nor a published SMC-specific recipe from [32].

The CRF row uses madmom’s historically named CRF-Beat DetectionProcessor [7] at 30–300 BPM with interval width 0.18; it should not be confused with the learned CRF in [8]. CombFilter uses madmom’s global tempo/iterative-alignment processor [12]. CRF and CombFilter decode downbeat activations at 8–100 BPM and snap their outputs to their respective beat grids.

#### 4.3. Frozen operating point and scale analysis

Every CASM row in Sec. 5 uses exactly one explicitly loaded Frozen-4F configuration:

$$\begin{aligned} F &= 50, \quad \theta_c = 0.03, \quad (B_{\min}, B_{\max}) = (30, 300), \\ |\mathcal{W}| &= 8 \text{ s}, \quad \beta = 0.15, \quad (T, L) = (2, 6), \\ (\lambda, \sigma_0, \sigma_u) &= (4, 0.12, 0.40), \quad [r_{\min}, r_{\max}] = [0.85, 1.8], \\ \mathcal{M} &= \{2, \dots, 7\}, \quad (T_d, \eta, \rho, \gamma) = (4, 3, 0, 0.6). \end{aligned}$$

Its scalar settings were selected on the union of SMC folds  $\{2, 7, 1, 3\}$  and then reused for all tracks and all backbones. Thus context-aware means input-conditioned inference, not test-time parameter updates: CASM has validation-selected parameters, but no annotated beat, inter-beat-interval, BPM, or meter input at inference.

The choice to foreground 4F was made after inspecting the calibration sensitivity analysis and is therefore exploratory. The seven-fold configuration is the only comparator locked before GTZAN inspection; it differs from 4F only in  $\sigma_0 = 0.15$ . For the scale study, SMC fold 0 ( $N = 27$ ) is never used for selection. From folds 1–7 we enumerate every 1-, 2-, and 4-fold calibration subset (7, 21, and 35 combinations) plus the seven-fold union, and evaluate all locked configurations on the same SMC fold 0 and fixed GTZAN/final1 panel. The latter panel is test-conditioned and is used only to describe sensitivity, not to claim a held-out optimum.

## 5. RESULTS

### 5.1. Main Result

Table 1 is best read as a transfer and trade-off table, not a winner count. At its displayed precision, the same CASM setting improves on Direct across the three backbones, two corpora, and all available metrics. Its characteristic behaviour is

Table 1. Comparison of beat-tracking post-processors across different backbone models. DBN default  $bpm = [55, 215]$  is for general music, but many SMC songs out of this range [2]; therefore, SMC-optimized  $bpm = [30, 300]$  can be applied. Only BPM setups were changed in post-processors, with other default setups unchanged.

| Method                | BPM    | GTZAN (Beat / Downbeat) |      |      |      |       |       | SMC (Beat) |      |      |
|-----------------------|--------|-------------------------|------|------|------|-------|-------|------------|------|------|
|                       |        | F1                      | CMLt | AMLt | F1   | CMLt  | AMLt  | F1         | CMLt | AMLt |
| BeatThis (direct) [4] | –      | 89.1                    | 79.8 | 89.8 | 78.3 | 67.2  | 79.1  | 62.7       | 51.4 | 61.0 |
| w. CASM               | 30–300 | +0.3                    | +0.8 | +1.2 | +0.7 | +3.6  | +8.0  | +0.3       | +2.4 | +3.6 |
| w. CASM, ablation     | 55–215 | +0.1                    | +0.5 | +0.6 | +0.5 | +2.9  | +5.8  | +0.1       | +0.4 | +2.0 |
| w. DBN [1]            | 30–300 | –0.5                    | –0.9 | +0.2 | –1.5 | –0.6  | +1.1  | –1.6       | +0.4 | +3.5 |
| w. DBN, default       | 55–215 | –1.0                    | +0.7 | +1.3 | –0.9 | +6.2  | +8.7  | –5.2       | –3.9 | +1.5 |
| w. PLPDP [5]          | 30–300 | +0.1                    | +0.3 | +0.3 | +0.1 | +0.2  | +0.2  | –0.9       | –3.5 | –0.3 |
| w. CombFilter [12]    | 30–300 | –4.0                    | –3.6 | +0.1 | –3.3 | +3.2  | +7.6  | –7.1       | +1.1 | +3.1 |
| w. CRF [7]            | 30–300 | –0.6                    | –0.3 | +1.4 | –1.4 | +1.9  | +5.5  | –1.2       | +1.4 | +1.2 |
| MSCNN (direct) [31]   | –      | 86.1                    | 73.4 | 79.9 | 70.8 | 53.1  | 63.1  | 57.2       | 42.0 | 48.4 |
| w. CASM               | 30–300 | +0.4                    | +1.6 | +2.4 | +1.5 | +12.5 | +18.3 | +0.3       | +3.5 | +6.8 |
| w. DBN                | 30–300 | –1.1                    | +0.2 | +5.9 | –0.9 | +9.5  | +14.6 | –0.2       | +2.2 | +4.1 |
| w. DBN, default       | 55–215 | –0.6                    | +4.0 | +7.2 | +1.8 | +15.8 | +21.2 | –4.9       | +0.3 | +3.9 |
| TCN (direct) [32]     | –      | 87.1                    | 73.8 | 81.7 | 62.4 | 24.6  | 59.9  | 52.9       | 30.1 | 36.0 |
| w. CASM               | 30–300 | +0.3                    | +1.3 | +1.6 | +2.1 | +9.2  | +13.4 | +1.7       | +4.5 | +8.2 |
| w. DBN                | 30–300 | –1.9                    | –0.8 | +4.4 | –0.2 | +8.9  | +9.3  | –0.4       | +2.1 | +5.9 |
| w. DBN, default       | 55–215 | –2.6                    | –0.7 | +5.0 | +4.9 | +23.2 | +22.9 | –0.9       | –7.0 | +2.3 |

conservative: it preserves local event placement while recovering part of the continuity absent from peak picking. DBN produces larger continuity gains in several GTZAN downbeat conditions, but these are often accompanied by lower beat or downbeat F1 and are less stable on SMC. The SMC-adjusted DBN likewise changes its trade-off when transferred back to GTZAN. We therefore do not claim that CASM dominates DBN, CRF, comb filtering, or PLPDP. The evidence instead supports a reusable operating point whose correction strength depends on the input rather than a new target-specific recipe.

### 5.2. What calibration-scale sensitivity shows

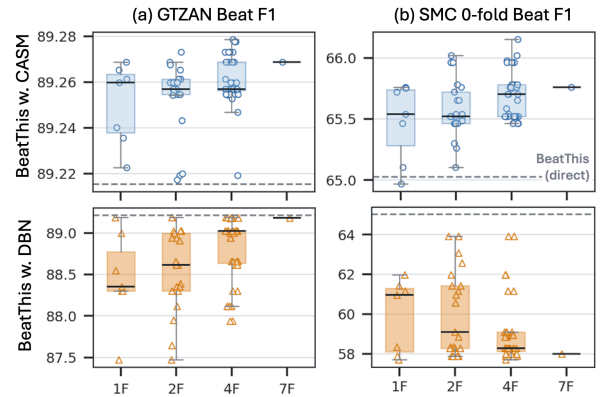


Fig. 3. Here is explain how to train Semi-Markov... about the data scaling problem

The fixed-panel sensitivity analysis does not show that 4F is optimal, nor that more data hurts machine learning. CASM learns no network weights here: the independent variable is the amount of data used to select global decoder scalars. On the held-out SMC panel, the family centres of F1, CMLt, and AMLt rise from 1F to 7F. On GTZAN, beat F1 and downbeat F1/CMLt peak at 2F, while beat

Table 2. Comparison with recent state-of-the-art beat tracking systems on GTZAN (mean over 3 seeds) and SMC dataset (mean over 8-fold). We only produced the bottom two lines, while the rest results are reported from their original works.

| Method                     | GTZAN (Beat / Downbeat) |      |      |      |      |      | SMC (Beat) |      |      |
|----------------------------|-------------------------|------|------|------|------|------|------------|------|------|
|                            | F1                      | CMLt | AMLt | F1   | CMLt | AMLt | F1         | CMLt | AMLt |
| HN-MERT w. DBN [34]        | 89.7                    | 81.4 | 94.3 | 77.4 | 71.6 | 87.2 | 61.7       | —    | —    |
| HN-MusicFM w. DBN          | 89.2                    | 80.9 | 93.7 | 79.8 | 73.2 | 89.5 | 60.6       | —    | —    |
| BeatFM-MERT w. DBN [35]    | 89.5                    | 82.7 | 93.9 | 76.7 | 70.2 | 85.4 | 61.3       | —    | —    |
| BeatFM-MusicFM w. DBN      | 89.1                    | 80.6 | 93.5 | 79.6 | 74.4 | 88.7 | 60.5       | —    | —    |
| BeatMamba w. DBN [36]      | 88.9                    | 82.3 | 94.3 | —    | —    | —    | 58.7       | 46.8 | 62.4 |
| BeatKAN w. DBN [37]        | 88.2                    | 78.1 | 92.3 | —    | —    | —    | 59.8       | 48.4 | 64.0 |
| MDM-BeatThis (direct) [27] | 89.7                    | 82.9 | 92.5 | 79.5 | 76.4 | 88.5 | 61.0       | 48.9 | 64.3 |
| BeatThis (direct) [4]      | 89.1                    | 79.8 | 89.8 | 78.3 | 67.2 | 79.1 | 62.7       | 51.4 | 61.0 |
| BeatThis w. CASM           | 89.4                    | 80.6 | 91.0 | 79.0 | 70.8 | 87.1 | 63.0       | 53.8 | 64.6 |

Following the evaluation protocols of prior work [4, 34, 35], we report the mean over three random seeds for GTZAN and the median over eight folds for SMC. For compactness, the table presents only these aggregated scores; complete seed- and fold-level results are provided in our GitHub repository. For CASM, the backbone predictions are generated in an eight-fold OOF setting, while the decoder configuration is selected using folds {2, 7, 1, 3}; therefore, this row should not be interpreted as strict nested decoder-selection OOF. CASM uses the Frozen-4F configuration selected using folds {2, 7, 1, 3}.

CMLt/AMLt and downbeat AMLt continue to 7F. Hence the defensible conclusion is metric-dependent diminishing returns rather than a universal best scale.

There is nevertheless a useful stability result. From 1F to 2F to 4F, the across-combination population spread decreases in all nine panels: more calibration data makes the selected decoder less sensitive to which SMC folds are present, even when the external point estimate does not improve monotonically. We retain 4F as a balanced, post-hoc operating point for the present analysis. Its near-identical seven-fold comparator is the appropriate pre-GTZAN locked reference; a new external test set or nested selection is needed to establish superiority.

### 5.3. Limitations and decisive next tests

Three limitations bound the current claim. First, SMC activations are backbone-level OOF, but the single 4F decoder is not selected in a nested outer loop. Second, CASM searches 30–300 BPM while the default DBN searches 55–215 BPM. Recent SMC analysis reports many references below that lower limit [2]; a matched-support DBN is therefore required before attributing an SMC difference solely to CASM’s ambiguity mechanism. Third, the present aggregate tables do not isolate the sparse graph, strength-versus-width ambiguity gate, or the two safeguards. A direct PLPDP baseline, fixed-period DP, component ablations, and per-corpus fall-back and meter-path acceptance rates are the decisive experiments. Track-paired bootstrap or randomization tests should accompany them; three training seeds or eight folds alone do not justify significance claims.

## 6. CONCLUSION AND FUTURE WORK

CASM revisits structured beat decoding as an inductive bias conditioned on each input, not as a return to a fixed global

grid. It joins activation-supported events with locally centred segment potentials and deliberately relaxes those potentials when periodic evidence is ambiguous. Once globally calibrated, the same deterministic module runs without upstream retraining, user annotation, or per-track tempo and meter settings. Current evidence supports a conservative F1–continuity alternative to Direct and DBN decoding, not universal dominance. The next version should use strict nested or leave-one-dataset-out calibration, matched tempo-support baselines, PLPDP and mechanism ablations, and explicit safeguard statistics. Multi-hypothesis period paths, causal decoding, and a differentiable duration layer are natural extensions, as is evaluation on every recent tracker that releases compatible activations.

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